

SCOT: Multi-Source Cross-City Transfer with Optimal-Transport Soft-Correspondence Objectives

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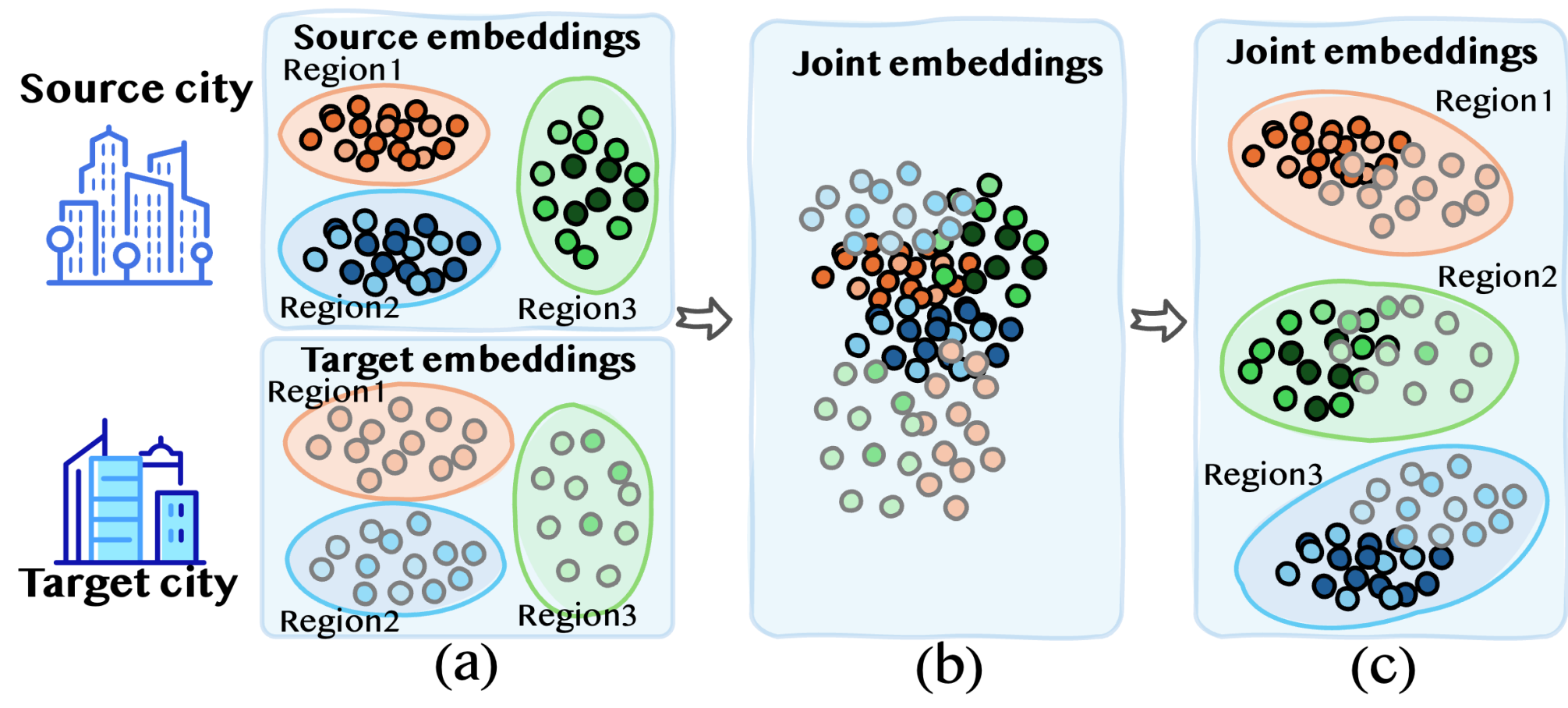
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1 Motivation

Labels (GDP, population, CO₂) exist only in a few well-instrumented cities. **How can we transfer them to label-scarce cities?**



Why is it hard?

No correspondence

Partitions differ across cities

$n_s \neq n_t$

Unequal, rectangular matching

Partial transfer

Corridors ✓ Tourist spots ✗

Key insight: the bottleneck is the **alignment**, not the encoder.

< 3%

encoder / readout swap

VS

25-40%

alignment design

Anchor / NN

RP · HBP · HSA

✗ Hubness

many-to-one collapse

Distribution

MMD · DANN

✗ Over-mixing

regions blur together

Soft OT (SCOT)

Sinkhorn + contrastive

✓ Selective

mass-controlled, many-to-many

Missing ingredient: explicit, mass-controlled **soft correspondence** between unequal region sets → **Optimal Transport**

Problem Setup & Data

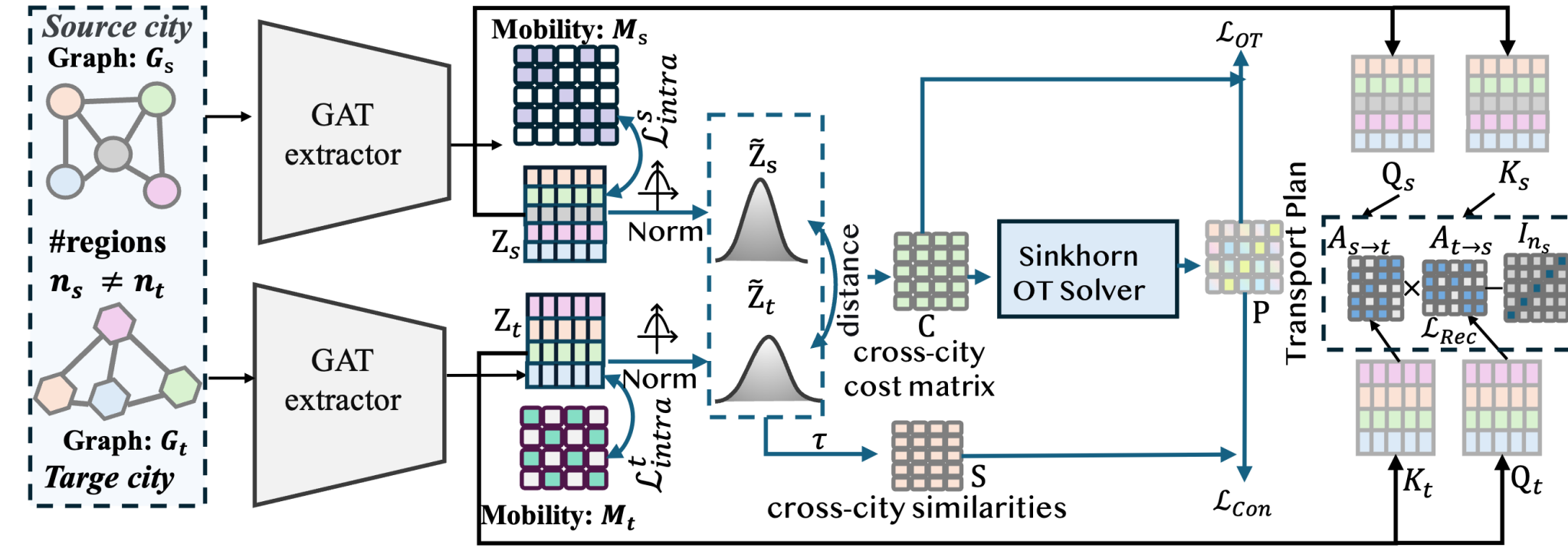
City	#Regions	#Trips	Mobility	Goal
Xi'an	1,306	559,729	$M_{ij} = \frac{\text{count}(i \rightarrow j)}{\sum_k \text{count}(i \rightarrow k)}$	$\mathbf{z}_s \in \mathbb{R}^{n_s \times d}$, $\mathbf{z}_t \in \mathbb{R}^{n_t \times d}$, $n_s \neq n_t$
Chengdu	1,056	384,618		
Beijing	1,311	78,945		
New York				

Tasks: region GDP · population · CO₂ (MAE / MAPE). No region correspondence is ever used.

Contributions

- OT soft correspondence: Sinkhorn OT + OT-weighted contrastive + one-sided cycle.
- Shared prototype hub: target-induced prior + balanced OT; no source selection.
- Consistent gains: 5-50% lower MAE/MAPE across cities, tasks & countries.

2 Method: SCOT



1 Intra-city encoding

→ preserve mobility

$$\mathbf{z}_c = \text{GAT}(\mathbf{H}_c^{(0)}; \mathbf{A}_c), \quad \hat{P}_{ij}^{(c)} = \frac{\exp(\mathbf{z}_{c,i}^\top \mathbf{z}_{c,j})}{\sum_k \exp(\mathbf{z}_{c,i}^\top \mathbf{z}_{c,k})}$$

$$\mathcal{L}_{\text{intra}} = - \sum_{c \in \{s, t\}} \sum_{i, j} (\mathbf{M}_c)_{ij} \log \hat{P}_{ij}^{(c)}$$

2 Sinkhorn soft correspondence

→ no hubness

$$\begin{aligned} C_{ij} &= \|\tilde{\mathbf{z}}_i^* - \tilde{\mathbf{z}}_j^*\|_2, \quad \mathbf{K} = \exp(-\mathbf{C}/\varepsilon) \\ \mathbf{u} &\leftarrow \mathbf{a} \odot (\mathbf{K}\mathbf{v}), \quad \mathbf{v} \leftarrow \mathbf{b} \odot (\mathbf{K}^\top \mathbf{u}) \quad (T \text{ iterations}) \\ \mathbf{P} &= \text{diag}(\mathbf{u}) \mathbf{K} \text{diag}(\mathbf{v}), \quad \mathcal{L}_{\text{OT}} = \frac{\langle \mathbf{P}, \mathbf{C} \rangle}{\min(n_s, n_t)} \end{aligned}$$

3 OT-weighted contrastive

→ no over-mixing

$$S_{ij} = \tilde{\mathbf{z}}_i^{s\top} \tilde{\mathbf{z}}_j^t / \tau, \quad \mathcal{L}_{\text{Con}} = - \frac{1}{n_s} \sum_{i=1}^{n_s} \log \frac{\sum_j P_{ij} e^{S_{ij}}}{\sum_j e^{S_{ij}}}$$

4 One-sided cycle regularizer

→ stability

$$\mathbf{A}_{s \rightarrow t} = \text{softmax}\left(\frac{\mathbf{Q}_s \mathbf{K}_t^\top}{\sqrt{d}}\right), \quad \mathcal{L}_{\text{Rec}} = \|\mathbf{A}_{s \rightarrow t} \mathbf{A}_{t \rightarrow s} - \mathbf{I}_{n_s}\|_F^2 + \beta \mathcal{R}_{\text{ent}}$$

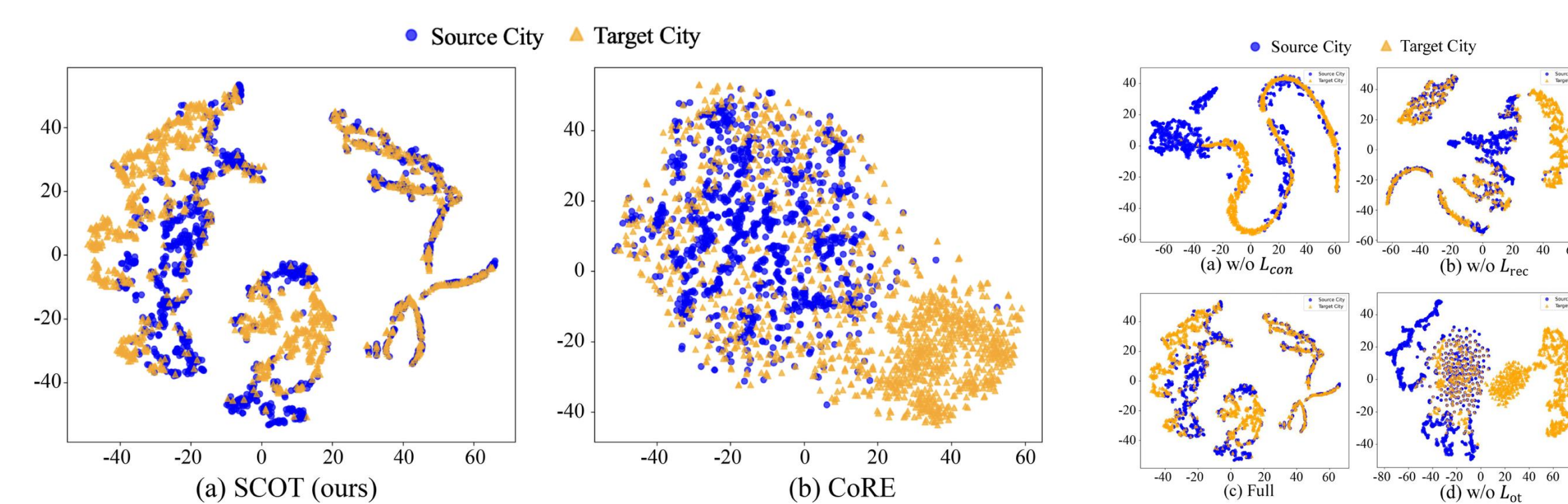
$$\begin{aligned} \mathcal{L} &= \mathcal{L}_{\text{intra}}^{(s)} + \mathcal{L}_{\text{intra}}^{(t)} + \lambda_{\text{align}} (\mathcal{L}_{\text{OT}} + \eta \mathcal{L}_{\text{Con}}) + \lambda_{\text{rec}} \mathcal{L}_{\text{Rec}} \\ \varepsilon &= 0.15, \quad \tau = 0.1, \quad \eta = 0.5, \quad \beta = 0.05, \quad \lambda_{\text{align}} = 1, \quad \lambda_{\text{rec}} = 0.5 \end{aligned}$$

Proposition: lower contrastive loss $\Rightarrow$ tighter transfer bound (empirical $r = 0.95$)

$$\begin{aligned} \mathcal{R}_t^b(h) &\leq \mathcal{R}_s^a(h) + (L_h + L_g) \sqrt{2 - 2\bar{m}} \\ \bar{m} &= \max \left\{ -1, \tau \log n_t + \tau H(\mathbf{a}) - \tau \mathcal{L}_{\text{Con}}(\mathbf{P}) - 1 - \frac{1}{2\tau} \right\} \end{aligned}$$

What SCOT learns: Aligned, not collapsed

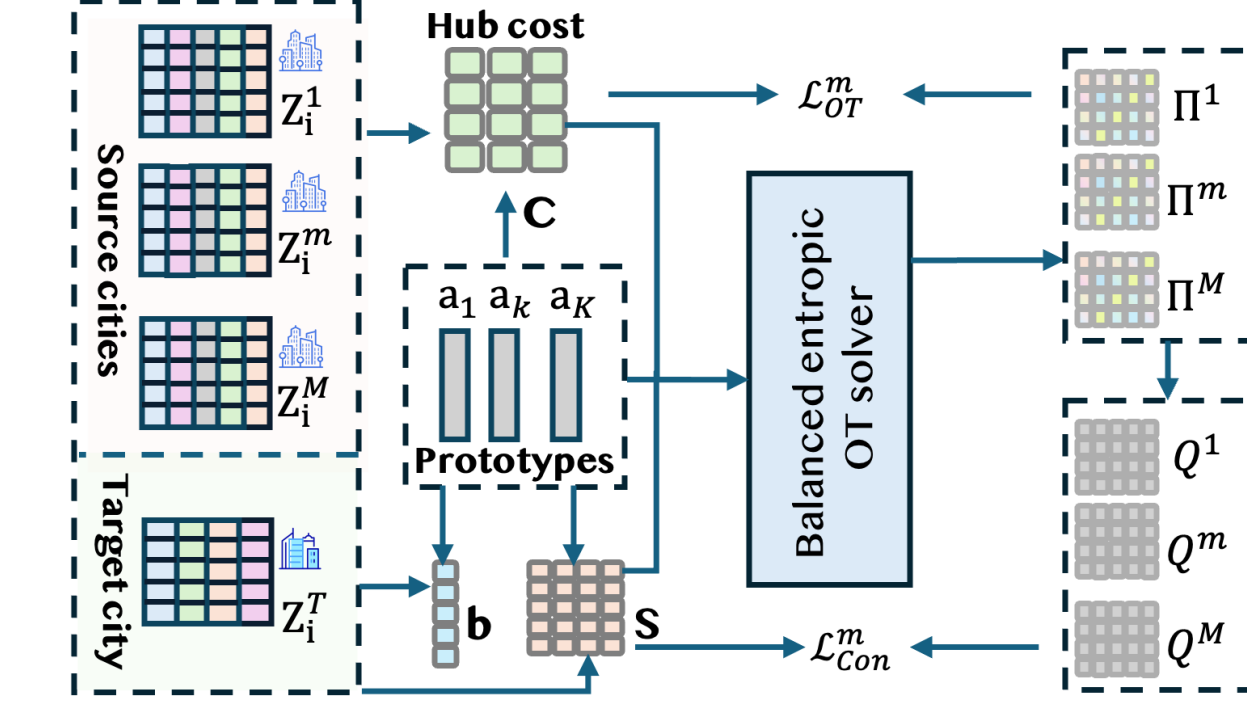
Every loss matters



SCOT vs. CoRE: cities interleave; CoRE leaves target clusters unmatched. 1.49× semantic lift.

Ablation: w/o $\mathcal{L}_{\text{con}}$ (a), $\mathcal{L}_{\text{rec}}$ (b), $\mathcal{L}_{\text{OT}}$ (d); full (c) aligns best.

3 Multi-Source Hub



- K = 32 shared prototypes as common anchors
- Balanced OT $\rightarrow$ no source domination
- Target-induced prior steers the hub
- ~30× cheaper than pairwise OT

$$\begin{aligned} \text{Target prior} \quad \bar{s}_k &= \frac{1}{n_t} \sum_{j=1}^{n_t} \tilde{\mathbf{z}}_j^t \tilde{\mathbf{a}}_k, \quad b_k \propto \max \{ \exp(\bar{s}_k / \tau_b), \epsilon_b \} \\ \text{City} \rightarrow \text{hub OT} \quad \Pi^m &= \arg \min_{\mathbf{P} \mathbf{1} = \mathbf{a}^m, \mathbf{P}^\top \mathbf{1} = \mathbf{b}} \langle \mathbf{P}, \mathbf{C}^m \rangle - \varepsilon H(\mathbf{P}), \quad C_{ik}^m = \|\tilde{\mathbf{z}}_i^m - \tilde{\mathbf{a}}_k\|_2 \\ \text{Objective} \quad \mathcal{L} &= \sum_{m \in \mathcal{S} \cup \{t\}} \mathcal{L}_{\text{intra}}^m + \frac{1}{|\mathcal{S}| + 1} \sum_m (\lambda_{\text{align}} \mathcal{L}_{\text{Align}}^m + \lambda_{\text{rec}} \mathcal{L}_{\text{Rec}}^m) \end{aligned}$$

4 Results

18/18

best MAE & MAPE (single-source)

-54.5%

max MAPE ↓ vs. best baseline (multi-source)

+21-32%

MAE gain with NYC as target

+3.9%

3-source hub vs. best single ($\rightarrow$ NYC)

Baselines (8 methods) grouped by alignment paradigm

Correspondence-based	RP · HBP · HSA — surrogate anchors (rank / hierarchical prototypes) + Procrustes / affine map
Correspondence-free	MMD · Adv (DANN) · CrossTreS (region re-weighting) · CoRE (global + region-level latent alignment)
Reference	Non-Alignment — intra-city objectives only, no cross-city signal

MAE ↓ vs. best baseline (6 directions)

Direction	GDP	Pop.	CO ₂
XA → BJ	-26.9%	-10.2%	-9.8%
BJ → XA	-4.5%	-9.5%	-9.1%
XA → CD	-4.8%	-6.6%	-8.2%
CD → XA	-8.9%	-10.4%	-5.8%
BJ → CD	-8.2%	-8.6%	-4.3%
CD → BJ	-9.1%	-8.2%	-2.0%

Single-source MAE, XA ↔ BJ red: best · blue: runner-up

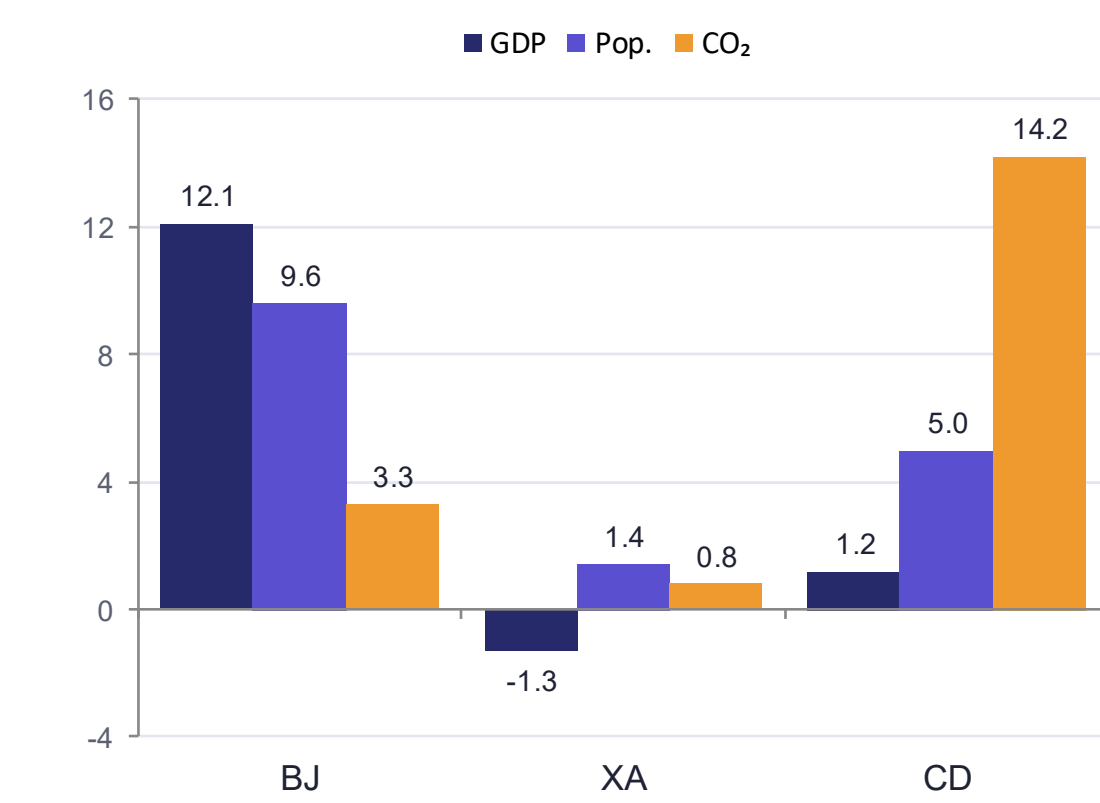
Method	GDP	Pop.	CO ₂	GDP	Pop.	CO ₂
Non-Align	264.30	981.07	288.41	252.91	946.93	270.42
RP	189.84	684.50	196.05	181.08	670.44	191.85
HBP	177.69	665.46	188.72	196.20	627.46	176.35
HSA	201.27	619.33	176.40	188.40	636.33	182.91
MMD	183.32	588.34	165.70	180.73	499.94	141.57
Adv	192.59	702.19	199.23	199.21	805.01	203.27
CrossTreS	207.43	633.25	179.75	170.27	639.44	182.72
CoRE	157.83	611.18	166.28	162.19	547.74	153.63
SCOT (ours)	115.33	528.50	149.42	154.92	452.67	128.74

Multi-source MAE (two sources → target) shared hub vs. baselines with adaptive source weighting

Method	Target: Beijing			Target: Xi'an			Target: Chengdu		
	GDP	Pop.	CO ₂	GDP	Pop.	CO ₂	GDP	Pop.	CO ₂
RP	172.76	679.86	166.80	195.91	642.01	181.16	187.13	630.44	167.76
HBP	164.25	662.60	165.53	200.04	670.41	150.16	176.35	631.26	125.97
HSA	156.40	644.89	160.05	183.55	648.67	155.20	180.71	651.06	159.86
MMD	127.45	605.81	160.52	163.78	506.22	144.61	163.82	639.32	145.74
Adv	196.76	717.96	189.41	221.31	731.92	184.73	183.26	687.72	156.37
CrossTreS	151.17	666.74	187.59	179.01	625.63	151.22	167.53	668.00	139.22
CoRE	152.88	620.34	152.24	173.72	549.37	134.19	156.10	621.11	121.33
SCOT (ours)	104.16	525.10	143.53	156.94	446.13	127.66	133.94	546.82	98.43

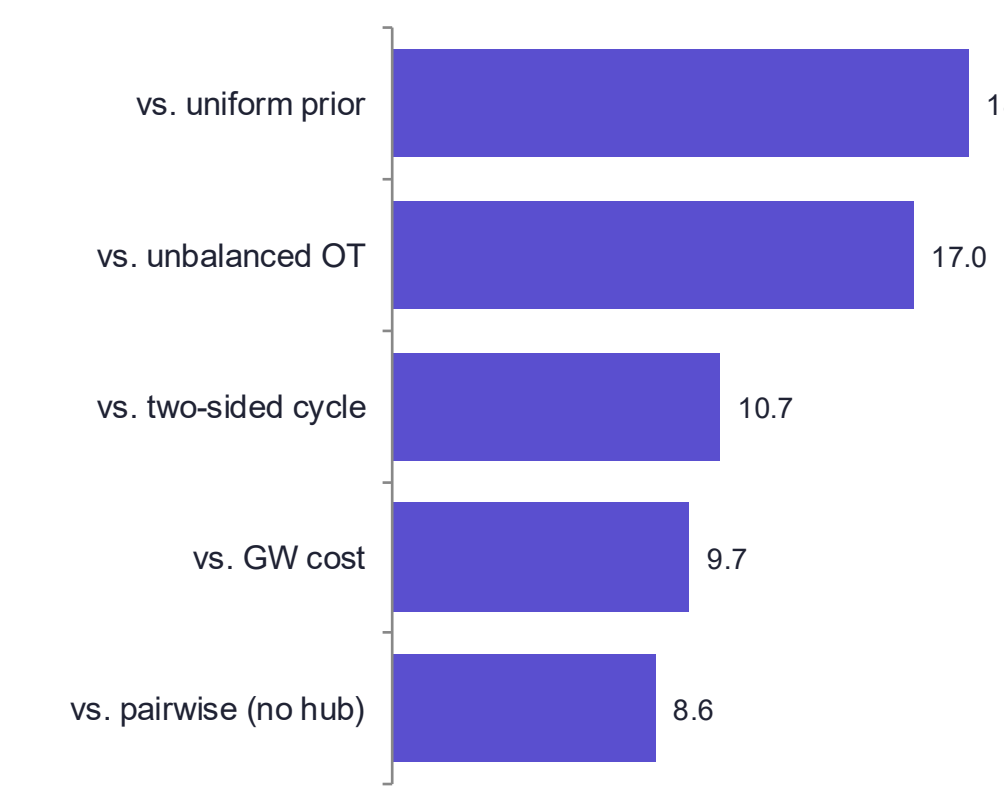
Multi- vs. best single-source

MAE reduction (%), per target



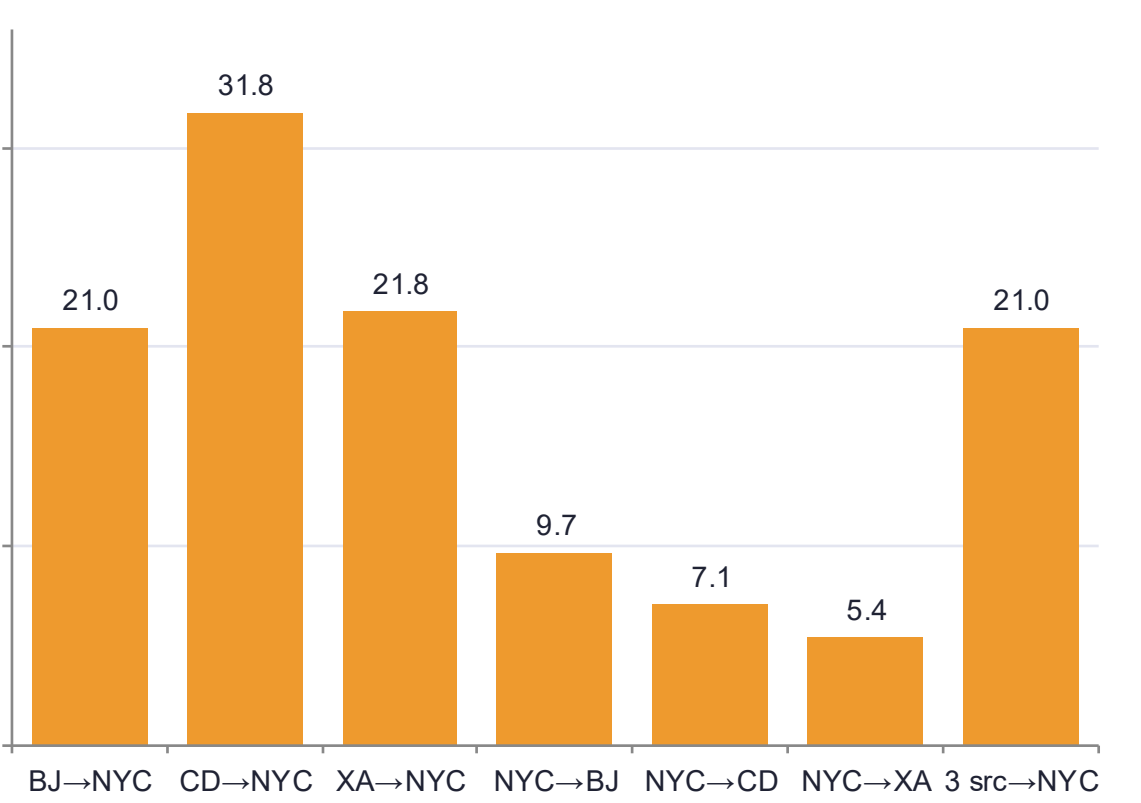
Design ablations

Pop. MAE reduction vs. alternative (%)



Cross-country: New York

Pop. MAE gain vs. strongest baseline (CoRE, %)



Diagnostics & robustness

35 / 1311

targets hold 50% of each row's OT mass

≈ 4

active prototypes per region (hub)

63%

OT mass between same-function blocks

1.49×

GT correspondence lift (synthetic, OT-only = 1.0)

< 3%

MAE change across encoders & readouts

Takeaways

- Invest in alignment, not encoders: it drives cross-city transfer.
- OT + contrastive: selective many-to-many matches, no hubness or over-mixing.
- A shared hub lets multiple sources help each other.